

2920/205

VISUAL PROGRAMMING

November 2021

Time: 3 hours



THE KENYA NATIONAL EXAMINATIONS COUNCIL

DIPLOMA IN INFORMATION COMMUNICATION TECHNOLOGY

MODULE II

VISUAL PROGRAMMING

3 hours

INSTRUCTIONS TO CANDIDATES

*This paper consists of **EIGHT** questions.*

*Answer **FIVE** of the **EIGHT** questions in the answer booklet provided.*

Candidates should answer the questions in English.

This paper consists of 7 printed pages.

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

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Turn over

1. (a) Explain the function of each of the following controls in a Visual Basic Program:

(i) DriveListBox;

(ii) DirListBox;

(iii) FileListBox.

(6 marks)

(b) Differentiate between a *ListBox* and *ComboBox* controls as used in a Visual Basic Program.

(4 marks)

(c) Explain each of the following terms as used in a Visual Basic program array:

(i) index;

(ii) element.

(4 marks)

(d) Figure 1 shows a Visual Basic program design window used to compute the sum of two numbers. The program uses a sub procedure to sum the two numbers that are specified by the arguments. Write the program attached to Calculate button and the corresponding procedure to perform the task.

(6 marks)

The figure shows a standard Windows-style window titled 'Form1'. Inside the window, there are two text boxes stacked vertically. The first text box is preceded by the label 'First Number' and the second by 'second Number'. Below these two text boxes is a single button labeled 'Calculate'.

Figure 1

2. (a) Explain the use of each of the following *error handling* statement in a Visual Basic Programming language:

(i) On Error GoTo 0;

(ii) On Error Resume Next;

(iii) On Error GoTo line.

(6 marks)

(b) Distinguish between *isNull* and *EOF* properties of data control as used in a Visual Basic program.

(4 marks)

(c) Explain **two** circumstances that may necessitate the conversion of a code statement into a comment in a Visual Basic program.

(4 marks)

(d) The following is a function in a Visual Basic program intended to multiply two integer variables passed to it in form of arguments x and y. The function will not work as written.

```
Private Function Mulply(x As Integer, y As Integer) As Integer
    Mulply = x * y
    Return Mulply
End Function
```

Identify the errors and provide corrections.

(6 marks)

3. (a) Outline the function of each of the following command button events as used in a Visual Program.
- (i) MouseHover;
 - (ii) KeyPress;
 - (iii) MouseDown.
 - (iv) MouseUp. (4 marks)
- (b) Distinguish between a *Multiple Document Interface (MDI) window* and *Single Document Interface (SDI) window* as used in a Visual Basic program. (4 marks)
- (c) Convert each of the following algebraic expressions into their equivalent Visual Basic code statement.
- (i) $(A + B)^{M \times N} - K$
 - (ii) $\frac{A^2 + B^B}{DDE}$ (4 marks)
- (d) Figure 2 shows a flow chart used to solve a mathematical problem. Use it to answer the questions that follow.

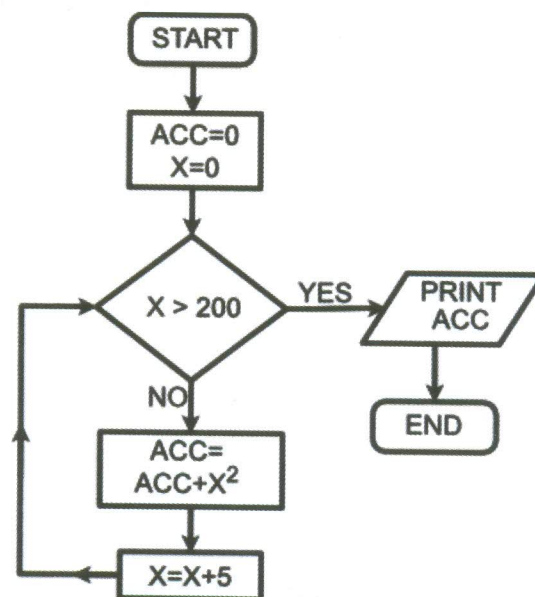


Figure 2

- (i) Explain **one** advantage of flow charts in programming. (2 marks)
- (ii) Convert the flow chart to its Visual Basic program equivalent. Use *do while*. (6 marks)

4. (a) Explain **two** types of cursors used in ActiveX Data Object control of a Visual Basic Program. (4 marks)
- (b) Explain a circumstance that may necessitate the use of each of the following keywords in a Visual Basic program:
- (i) Option Explicit;
- (ii) ReDim. (4 marks)
- (c) A student created a project in a Visual Basic program and opted to use stored procedures to query a database. Explain **two** benefits gained from using this procedure. (4 marks)
- (d) A Company management has opted to pay travel allowance for its workers based on salary earned as shown in table 1.

Salary	Travel Allowance
From 0 to 9999	2000
From 10000 to 19999	3300
From 20000 to 29999	4000
From 30000 to 60000	5000

Table 1

Write a Visual Basic program that allows a user to enter salary value in a text box named *txtIncome*. The program should then display the corresponding travel allowance on a label named *lblTravelAll*. If an invalid income value is entered, a message box should be displayed alerting the user that the salary value is not valid and the Travel Allowance set to -1. Use *Select Case* statement. (8 marks)

5. (a) Outline **three** uses of a *Project Explorer window* in a Visual Basic program. (3 marks)
- (b) Outline **three** limitations of using Visual Basic program to develop software applications. (3 marks)
- (c) Distinguish between each of the following pairs of terms as used in a Visual Basic program.
- (i) List Index and Tab index;
- (ii) Datafield and DataSource. (8 marks)
- (d) A teacher intends to compute the average scores obtained by a student in examinations and determine the scores that are less than the average.

Write a Visual Basic program that prompts for the scores of five students and assign the values to an array. The program should compute and output the average score and the scores which are less than the average on a form. Attach the code to a command button. (6 marks)

6. (a) Outline each of the following types of projects that can be created in Visual Basic program:
- (i) Standard EXE;
 - (ii) VB application wizard;
 - (iii) Data project;
 - (iv) Addin. (4 marks)
- (b) Distinguish between *relational operators* and *logical operators* as used in a Visual Basic program: (4 marks)
- (c) State the function of each of the following field values entered in a footer section of a Visual Basic program report:
- (i) %p
 - (ii) %P
 - (iii) %d
 - (iv) %t
 - (v) %i (5 marks)
- (d) Figure 3 shows a Visual Basic program design window having three *control array option buttons* named *activity*.

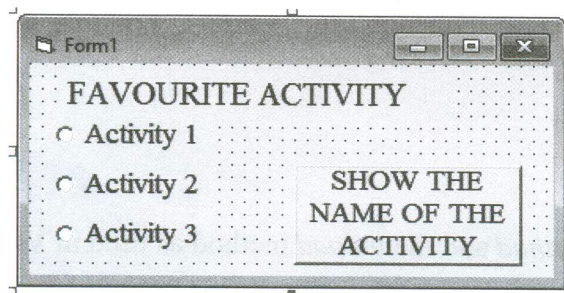


Figure 3

When a user selects an activity option and clicks a button, the corresponding activity name is displayed using a message box as shown in Table 2.

Button	Activity Name
Activity 1	Exotic car driving
Activity 2	Swimming
Activity 3	Dancing

Table 2

Write a program to display activity name for the corresponding option button selected. Use *select case*. (7 marks)

7. (a) Explain **two** types of *recordset* used in a Visual Basic program. (4 marks)
- (b) State the difference between Recordset *Find* methods and *Move* methods as used in Visual Programming. (4 marks)
- (c) The following is a Visual Basic Program code segment. Use it to answer the question that follow

```
F=0 J o 1 T= 10 S = 2
F=0 K o T= 5
X o X + 1
N+eK
N+eK
```

Rewrite this code with an equivalent "Do" loop structure. (5 marks)

- (d) The formula $F = \frac{9}{5}C + 32$ is used to convert temperature value from Celsius to Fahrenheit where F is temperature in Fahrenheit and C is temperature in Celsius.

Write a Visual Basic program that prompts a user to enter temperature in Celsius. The program should use a function procedure to convert the temperature to its equivalent in Fahrenheit and display the output on a picture box. (7 marks)

8. (a) Outline the function of each of the following control properties in a Visual Basic program:

(i) Appearance;

(ii) Back Color; - prior type that appears at the background

(iii) Left & Top;

(iv) Enabled. (4 marks)

- (b) Distinguish between the *hide method* and the *unload method* as used in Visual Programming. (4 marks)

- (c) Explain a reason for the occurrence of each of the following errors in a Visual Basic program:

(i) Division by zero;

(ii) Overflow;

(iii) Subscript out of range. (6 marks)

- (d) Write a Visual Basic program that would display the following pattern on the form. Use *while* loop. (6 marks)

```
X
XX
XXX
XXXX
XXXXX
XXXXXX
XXXXXXX
XXXXXXXX
XXXXXXXXX
XXXXXXXXXX
XXXXXXXXXX
```

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